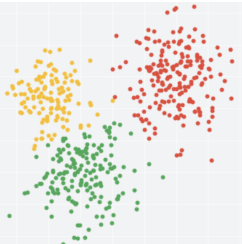


KMedoids

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Issues with k -Means algorithm

- ▶ Results can vary depending on initial random choices
 - ▶ Can get trapped in a local minimum that isn't the global optimal solution
 - ▶ Repeat the clustering procedure multiple times with different initialisations and select the *best* final clustering
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- ▶ Outliers have a larger effect on the mean value, hence cluster centre and the cluster
 - ▶ Cluster centres (means) are not actual instances in the cluster
 - ▶ Euclidean distance used in the algorithm is inappropriate for categorical features

k -Medoids algorithm

- ▶ Representative-based algorithms
- ▶ The goal is to determine k representatives $\bar{Y}_1, \dots, \bar{Y}_k$ that minimise the following objective function

$$\sum_{i=1}^n \left[\min_j d(\bar{X}_i, \bar{Y}_j) \right]$$

Unlike the k -Means, in the k -Medoids algorithm

- ▶ the **distance function** (dissimilarity measure) $d(\cdot, \cdot)$ can be any function convenient for the dataset (not necessarily the Euclidean distance)
- ▶ representatives are selected from the dataset

k -Medoids algorithm

Uses hill-climbing strategy:

1. Start with an arbitrary solution to a problem
2. Attempt to find a better solution by making an incremental change to the solution.
3. If the change produces a better solution, another incremental change is made to the new solution, and so on
4. Until no further improvements can be found.

k -Medoids algorithm

k -MedoidsClustering (Number of clusters: k , Dataset: $\mathcal{D} = \{\bar{X}_1, \dots, \bar{X}_n\}$)

1. Initialisation phase

Choose k cluster representatives (medoids) $\bar{Y}_1, \dots, \bar{Y}_k$ from the dataset randomly

2. Assignment phase

Assign all objects in the dataset to the closest representative.

The resulting clusters: $C_1, \dots, C_k$

3. Optimisation phase (hill-climbing step)

1. Find a pair $(\bar{X}, \bar{Y})$, where $\bar{X} \in \mathcal{D}$ and $\bar{Y} \in \{\bar{Y}_1, \dots, \bar{Y}_k\}$ such that
2. Replacing $\bar{Y}$ with $\bar{X}$ in the set of representatives leads to the greatest possible improvement in the objective function
3. If improvement is positive then replace $\bar{Y}$ with $\bar{X}$ and go to phase 2. Otherwise return current clusters $C_1, \dots, C_k$

k -Medoids algorithm

Pros

- ▶ Representatives are chosen from the dataset
 - ▶ allows for greater interpretability of the cluster representatives
 - ▶ more robust to noise and outliers than k -means
- ▶ Can be used with arbitrary dissimilarity measures
 - ▶ thus applicable to data of complex data type (categorical, mixed, time-series, etc.)

Cons

- ▶ Results can vary depending on initial random choices
- ▶ Can get trapped in a local minimum that isn't the global optimal solution
 - ▶ Repeat the clustering procedure multiple times with different initialisations and select the best final clustering
- ▶ Slower than the k -means algorithm

k -Medoids algorithm: time-complexity issue

If we have n objects in the dataset, then at each execution of the optimisation phase we need to compute $k \cdot n$ times the incremental objective-function change (this is too expensive).

3. Optimisation phase (hill-climbing step)

1. Find a pair $(\bar{X}, \bar{Y})$, where $\bar{X} \in \mathcal{D}$ and $\bar{Y} \in \{\bar{Y}_1, \dots, \bar{Y}_k\}$, such that
2. Replacing $\bar{Y}$ with $\bar{X}$ in the set of representatives leads to the greatest possible improvement in the objective function.
3. If the improvement is positive then replace $\bar{Y}$ with $\bar{X}$ and go to phase 2. Otherwise return the current clusters $C_1, \dots, C_k$.

A solution to this is to use a randomly selected set of r pairs $(\bar{X}, \bar{Y})$, where $\bar{X} \in \mathcal{D}$ and $\bar{Y} \in \{\bar{Y}_1, \dots, \bar{Y}_k\}$, and use the best of these pairs for the replacement. In this way we need to compute only r times the incremental objective-function change.